

MTH166

Lecture-26

**Boundary Value Problems of Heat
Equations**

Topic:

Solution of Partial Differential Equations

Learning Outcomes:

To solve Boundary value problems of one dimensional Heat Equation.

Recall: Heat Equation

Equation: $\frac{\partial u}{\partial t} = C^2 \frac{\partial^2 u}{\partial x^2}$

Nature: Parabolic

Solution: 1. $u(x, t) = Ax + B$

2. $u(x, t) = (Ae^{px} + Be^{-px}) e^{C^2 p^2 t}$

or

$$u(x, t) = (A \cosh px + B \sinh px) e^{C^2 p^2 t}$$

3. $u(x, t) = (A \cos px + B \sin px) e^{-C^2 p^2 t}$ **(Most suitable one)**

Problem. Solve $\frac{\partial u}{\partial t} = C^2 \frac{\partial^2 u}{\partial x^2}$, $0 \leq x \leq l, t > 0$ under boundary conditions:

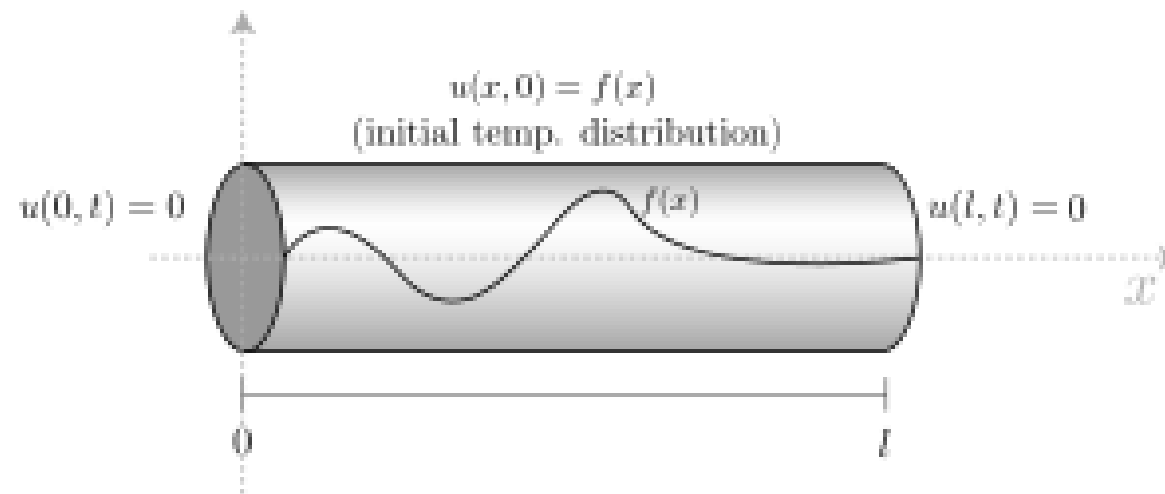
$$u(0, t) = 0, u(l, t) = 0.$$

Solution. The given one dimensional Heat equation is:

$$\frac{\partial u}{\partial t} = C^2 \frac{\partial^2 u}{\partial x^2} \quad (1)$$

$$0 \leq x \leq l \text{ (length)}, t > 0 \text{ (time)}$$

$$\text{Such that : } u(0, t) = 0, u(l, t) = 0.$$



We know the most appropriate solution of equation (1) is:

$$u(x, t) = (A \cos px + B \sin px)e^{-c^2 p^2 t} \quad (2)$$

Using the condition $u(0, t) = 0$, we get:

$$u(0, t) = (A \cos 0 + B \sin 0)e^{-c^2 p^2 t}$$

$$\Rightarrow 0 = A \cdot e^{-c^2 p^2 t} \quad \Rightarrow \mathbf{A = 0}$$

Equation (2) becomes:

$$u(x, t) = (B \sin px)e^{-c^2 p^2 t} \quad (3)$$

Using the condition $u(l, t) = 0$, we get:

$$u(l, t) = (B \sin pl)e^{-c^2 p^2 t}$$

$$\Rightarrow 0 = (B \sin pl)e^{-c^2 p^2 t}$$

$$\Rightarrow 0 = (B \sin pl)e^{-c^2 p^2 t}$$

$$\Rightarrow \sin pl = 0 \quad (\text{As } B \neq 0)$$

$$\Rightarrow \sin pl = \sin n\pi$$

$$\Rightarrow pl = n\pi \quad \Rightarrow p = \frac{n\pi}{l}$$

Equation (3) becomes:

$$u(x, t) = (B \sin px)e^{-c^2 p^2 t}$$

$$\Rightarrow u(x, t) = (B \sin \frac{n\pi x}{l})e^{-\frac{c^2 n^2 \pi^2}{l^2} t} \quad (4)$$

This gives temperature distribution only at one point. There are infinite number of points on the rod.

By principle of superposition:

$$u(x, t) = \sum_{n=1}^{\infty} (B_n \sin \frac{n\pi x}{l}) e^{-\frac{c^2 n^2 \pi^2}{l^2} t} \quad (5)$$

Which is the required solution of Heat equation (1).

$$\text{Where } B_n = \frac{1}{2l} \int_0^l u(x, 0) \sin \frac{n\pi x}{l} dx$$

Here $u(x, 0)$ is the initial displacement.

Polling Quiz

If $u(x, t) = (A \cos px + B \sin px)e^{-c^2 p^2 t}$ is the solution of one-dimensional heat equation, then which of the following is true when initial condition $u(0, t) = 0$ is applied to the solution:

(A) $A = 0$

(B) $p = n\pi/l$

(C) $B = 0$

(D) $C = 0$

